

Agentic AI in Cybersecurity

Learning about AI Agents and Use in Cybersecurity



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Course Introduction

Threat Detection and Prevention

- How AI detects anomalies
- AI capabilities and cybersecurity tools

Integration into Security Frameworks

- AI-human collaboration
- Automated incident response

Industry value

- Getting started with Agentic AI
- Industry use cases





What is Agentic AI

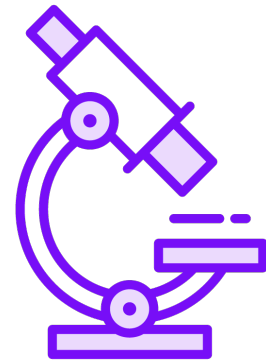
Agentic AI is a type of artificial intelligence that can make autonomous decisions and take goal directed actions based on its environment, much like an intelligent assistant that acts independently to solve problems.



Why Agentic AI Matters Now



Cyberattacks



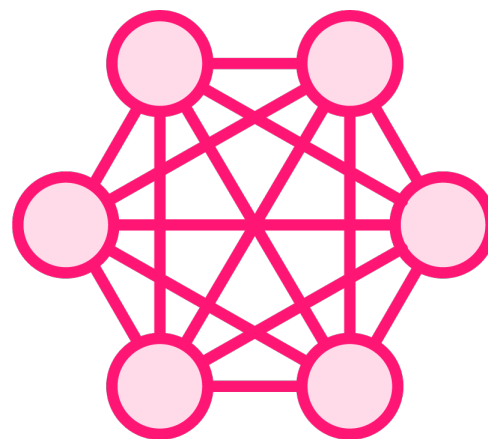
Sophisticated



**Overwhelmed by
Alerts**



Autonomous



Adaptive



Real Time



Agentic AI in Threat Detection

**Monitor behavior,
traffic, and patterns
continuously**

**Detects anomalies in
real time, not just
known threats**

**Learns what “normal”
looks like to spot
subtle changes**

**Reduces false
positives by focusing
on behavior context**

**Provides early
warning before
attacks escalate**



Traditional vs Agentic Tools

Traditional

- Rule based and signature driven**
- Reactive: responds after an alert**
- High false positive rate**
- Static, needs regular manual updates**
- Overwhelms teams with alert volume**

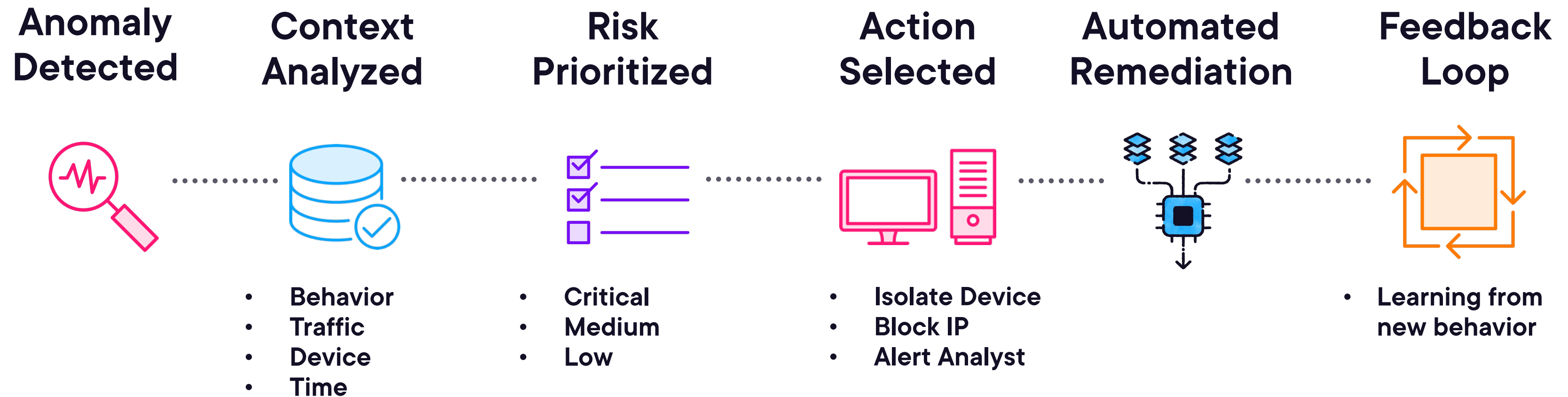
vs.

Agentic tools

- Context aware and behavior driven**
- Proactive: detects and acts in real time**
- Prioritizes and filters high-risk threats**
- Continuously learns and adapts**
- Automates response to reduce workload**



Prioritization & Remediation





Case Study: Ransomware Attack

Phishing email contains unknown ransomware

Traditional tools failed to flag new malware

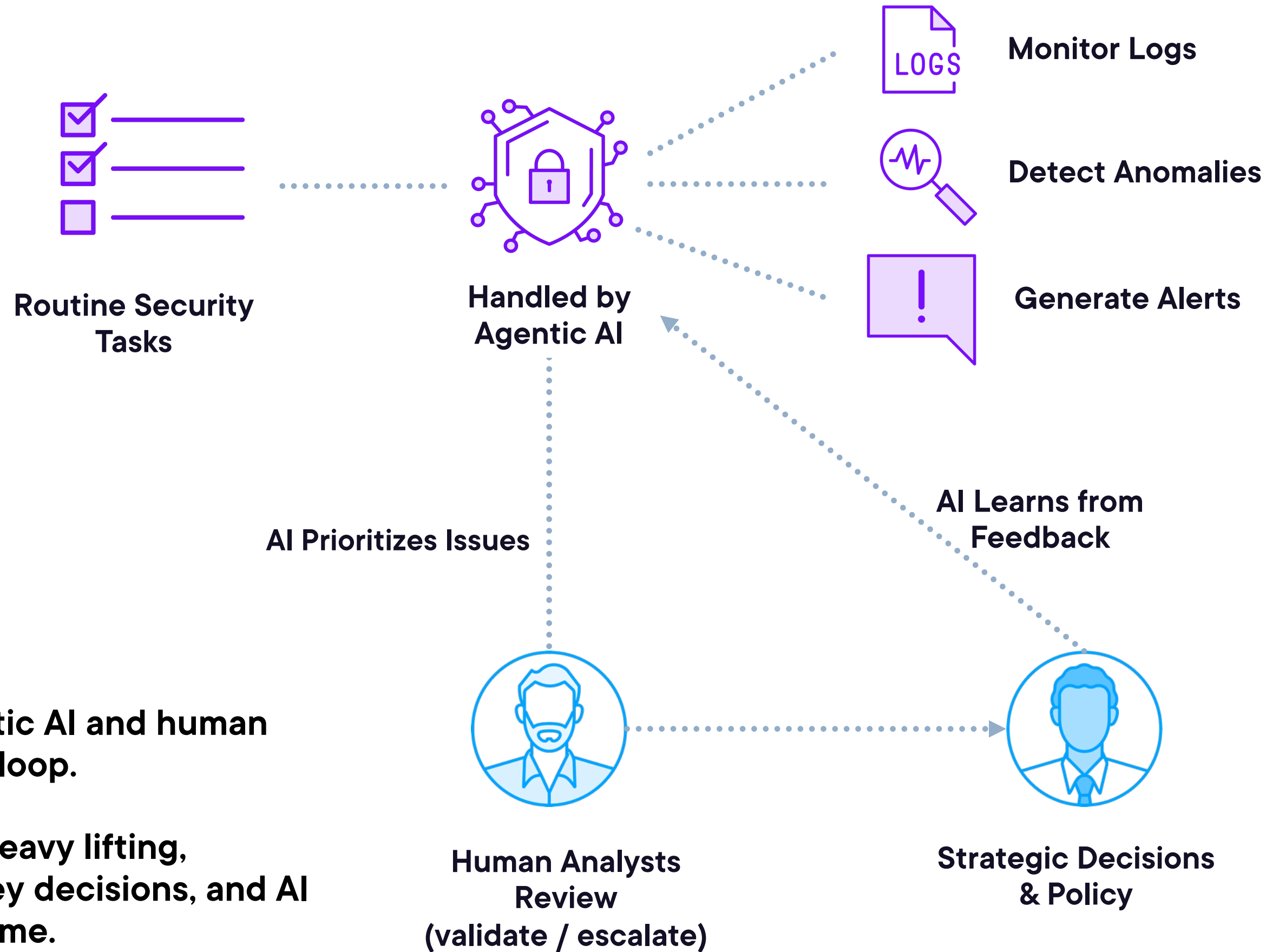
Agentic AI detected abnormal file encryption patterns

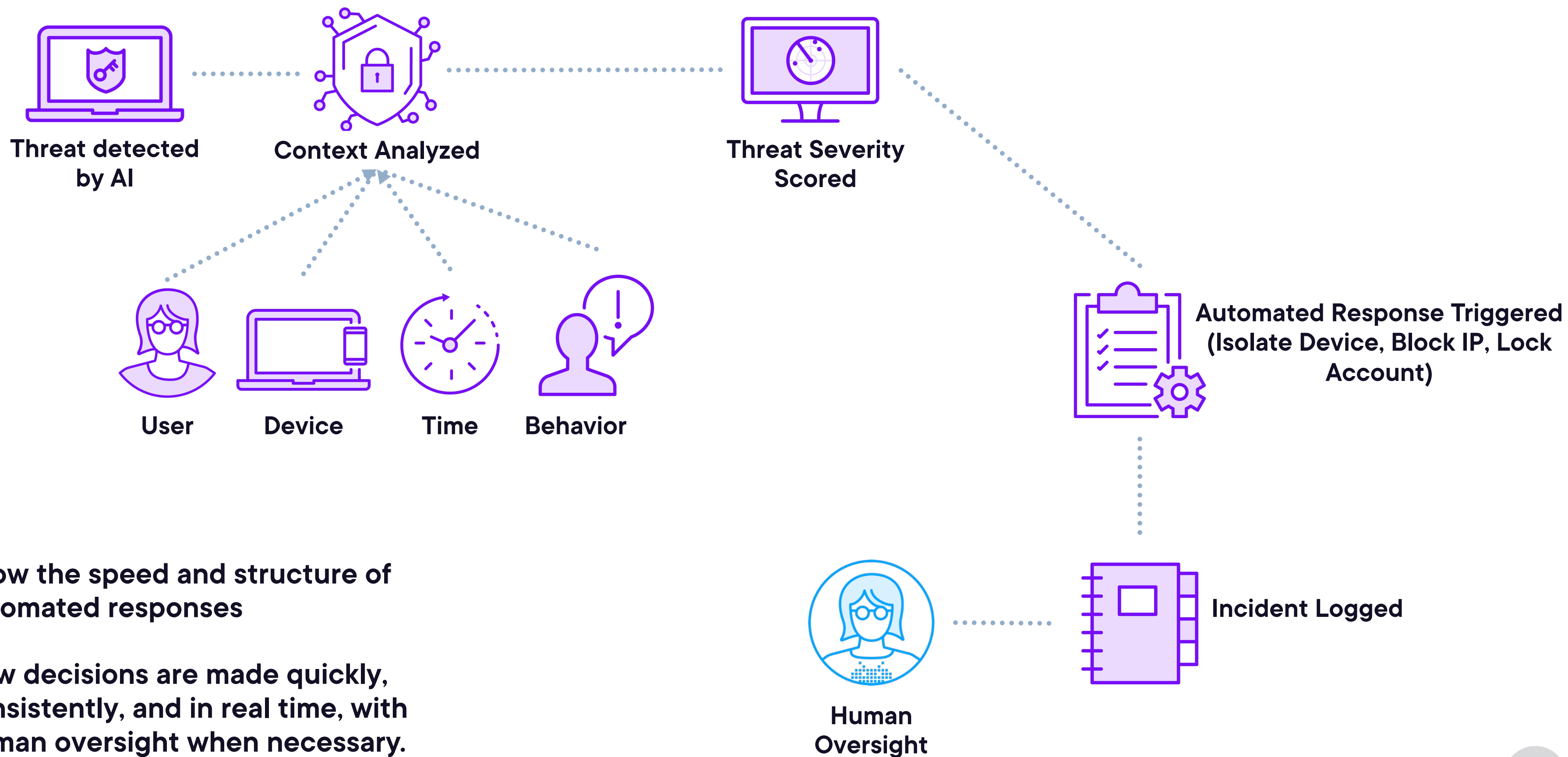
Compromised device was isolated automatically

Attack contained before it could spread

Millions in potential damage were prevented







Show the speed and structure of automated responses

How decisions are made quickly, consistently, and in real time, with human oversight when necessary.



Continuous Learning

Learns from every interaction

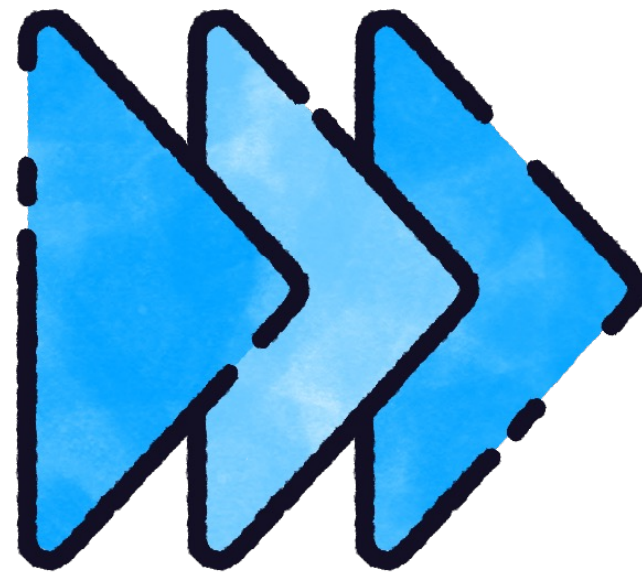
Adapts to evolving threats

Reduces false positives over time

**Strengthens your security posture
continuously**



Getting Started With Agentic AI



Where to begin:

- pilot projects
- augmenting SIEM/SOAR
- vendor selection

Tips for leaders:

- involve security teams
- define goals
- measure outcomes

3-step maturity model

- Start → Scale → Optimize



Use Cases for Healthcare

**Detects abnormal
access to patient
records in real time**

**Flags insider threats
based on unusual
user behavior**

**Monitors medical
devices for early
signs of ransomware**

**Automatically
isolates compromised
systems to prevent
spread**

**Supports compliance
and data protection
in high-risk
environments**



Use Cases for Finance

**Monitors transactions in real time
to detect fraud instantly**

**Freezes suspicious accounts
automatically to prevent loss**

**Identifies credential theft through
behavior and access anomalies**

**Protects trading systems and
ensures compliance without delay**



Use Cases for Retail

**Detects and blocks
credential stuffing
and bot attacks in
real time**

**Flags unusual POS
activity and internal
fraud automatically**

**Scales protection
during peak sales
without disrupting
customers**



Course Summary

What agentic AI is and how it differs from traditional automation.

Learn how agentic AI detects cyber threats using real-time behavioral analysis.

Explore how AI agents prioritize and remediate vulnerabilities automatically.

Compare traditional security tools with adaptive, agent-based approaches.

Discover real-world applications in healthcare, finance, and retail sectors.

Get practical steps for integrating agentic AI into your security strategy.



Additional Courses

Vlad Catrinescu, Using AI Agents for Productivity, [Using AI Agents for Productivity](#)

Jamie Maguire, Aligning Generative AI with Business Cases, [Aligning Generative AI with Business Cases](#)

Tom Taulli, Integrating AI Agents in Enterprise Systems, [Integrating AI Agents in Enterprise Systems](#)

Pratheerth Padman, Getting Started with Artificial Intelligence for Business, [Getting Started with Artificial Intelligence for Business](#)

